Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #8

[1] Prove that the best prediction of the outcome of a continuous random variable is its mean. Best is to be interpreted as the value that minimizes $mse(b) = E[(X-b)^2]$.

Solution: $mse(b) = E[(X-b)^2] = E[X^2 - 2bX + b^2] = E[X^2] - 2bE[X] + b^2$

$$\frac{dmse(b)}{db} = 0 - 2E[X] + 2b = 0 \Rightarrow b_{opt} = E[X], \quad \hat{x} = E[X]$$ ∎

[2] Determine the mean of the following PDFs

(a) X_N^2 PDF
(b) Erlang PDF
(c) Rayleigh PDF

Solution: (a) $$P_X(x) = \frac{1}{2^{\frac{N}{2}}\Gamma(\frac{N}{2})} x^{\frac{N}{2}-1} e^{-\frac{1}{2}x}, \quad 0 \le x < \infty$$

$$E[X] = \int_0^\infty x P_X(x)\,dx = \int_0^\infty x \cdot \frac{1}{2^{\frac{N}{2}}\Gamma(\frac{N}{2})} x^{\frac{N}{2}-1} e^{-\frac{1}{2}x}\,dx$$

$$= \frac{1}{2^{\frac{N}{2}}\Gamma(\frac{N}{2})} \left[\int_0^\infty \frac{1}{2^{\frac{N+2}{2}}\Gamma(\frac{N+2}{2})} x^{\frac{N+2}{2}-1} e^{-\frac{1}{2}x}\,dx \right] \cdot 2^{\frac{N+2}{2}} \cdot \Gamma\left(\frac{N+2}{2}\right)$$

$$= \frac{2\,\Gamma(\frac{N+2}{2})}{\Gamma(\frac{N}{2})} \int_0^\infty \underbrace{\frac{1}{2^{\frac{M}{2}}\Gamma(\frac{M}{2})} x^{\frac{M}{2}-1} e^{-\frac{1}{2}x}\,dx}_{X_M^2 \text{ PDF, Hence integral is equal to 1}} \qquad (M = N+2)$$

$$= \frac{2\,\Gamma(\frac{N+2}{2})}{\Gamma(\frac{N}{2})} = 2\,\frac{\Gamma(\frac{N}{2}+1)}{\Gamma(\frac{N}{2})} = \frac{2(\frac{N}{2})\Gamma(\frac{N}{2})}{\Gamma(\frac{N}{2})} = 2\frac{N}{2} = N$$

(b) $$P_X(x) = \frac{\lambda^N}{(N-1)!} x^{N-1} e^{-\lambda x}, \quad 0 \le x < \infty$$

$$E[X] = \int_0^\infty x P_X(x)\,dx = \int_0^\infty x \frac{\lambda^N}{(N-1)!} x^{N-1} e^{-\lambda x}\,dx = \frac{\lambda^N}{(N-1)!} \int_0^\infty x^{(N+1)-1} e^{-\lambda x}\,dx$$

$$= \frac{\lambda^N}{(N-1)!} \int_0^\infty \frac{N!}{\lambda^{N+1}} \cdot \frac{\lambda^{N+1}}{N!} x^{(N+1)-1} e^{-\lambda x}\,dx = \frac{\lambda^N}{(N-1)!} \frac{N!}{\lambda^{N+1}} \underbrace{\int_0^\infty \frac{\lambda^{N+1}}{N!} x^{(N+1)-1} e^{-\lambda x}\,dx}_{\text{Erlang PDF with parameter } N=N+1}$$

$$= \frac{\lambda^N}{(N-1)!} \frac{N!}{\lambda^{N+1}} = \frac{N}{\lambda}$$

(c) $P_X(x) = \frac{x}{\sigma^2} e^{\frac{-x^2}{2\sigma^2}}, \quad 0 \le x < \infty$

$$E[X] = \int_0^\infty x P_X(x)\,dx = \int_0^\infty x \cdot \frac{x}{\sigma^2} e^{\frac{-x^2}{2\sigma^2}} dx$$

$u = x, \quad dv = \frac{x}{\sigma^2} e^{\frac{-x^2}{2\sigma^2}} dx \Rightarrow v = -e^{-\frac{1}{2\sigma^2}x^2}$

$$E[X] = -x e^{\frac{-x^2}{2\sigma^2}}\Big|_0^\infty - \int_0^\infty -e^{\frac{-x^2}{2\sigma^2}} dx = 0 + \int_0^\infty e^{\frac{-x^2}{2\sigma^2}} dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} e^{\frac{-x^2}{2\sigma^2}} dx = \frac{1}{2}\cdot\sqrt{2\pi\sigma^2}\int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}}}_{N[0,\sigma^2]} dx$$

$$= \frac{1}{2}\sqrt{2\pi\sigma^2} = \sqrt{\frac{\pi\sigma^2}{2}}$$

[3] Determine $E[X^2]$ and $E[(2X+1)^2]$ for a $N[m,\sigma^2]$ random variable.

Solution: $\text{var}(X) = E[X^2] - (E[X])^2 \Rightarrow E[X^2] = \text{var}(X) + (E[X])^2 = \sigma^2 + m^2$

$$E[(2X+1)^2] = E[4X^2 + 4X + 1] = 4E[X^2] + 4E[X] + 1$$
$$= 4(\sigma^2 + m^2) + 4m + 1$$
$$= 4\sigma^2 + 4m^2 + 4m + 1$$

[4] A mixed PDF is given as $P_X(x) = \frac{1}{2}\delta(x) + \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}} u(x)$. What is $E[X^2]$?

Solution: $E[X^2] = \int_{-\infty}^{\infty} x^2 P_X(x)\,dx$

$$= \int_{-\infty}^{\infty} x^2\left[\frac{1}{2}\delta(x) + \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}} u(x)\right] dx = \int_{-\infty}^{\infty} \frac{x^2}{2}\delta(x)\,dx + \int_0^\infty \frac{x^2}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}} dx$$

$$= 0^2\cdot\frac{1}{2} + \frac{1}{2}\underbrace{\int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}} dx}_{E[X^2] \text{ for } N[0,\sigma^2]} = \frac{1}{2}[\sigma^2 + 0^2] = \frac{\sigma^2}{2}$$

[5] Find the mean and variance for a Laplacian random variable.

Solution: $P_X(x) = \frac{1}{\sqrt{2\sigma^2}} e^{-\sqrt{\frac{2}{\sigma^2}}|x|}, \quad -\infty < x < \infty$

$$E[X] = \int_{-\infty}^{\infty} x P_X(x)\,dx = \int_{-\infty}^{\infty} \underbrace{x}_{\text{odd function}} \underbrace{\frac{1}{\sqrt{2\sigma^2}} e^{-\sqrt{\frac{2}{\sigma^2}}|x|}}_{\text{even function}} dx = 0$$

since odd × even = odd and $\int_{-a}^{a} f(x)\,dx = 0$ if $f(x)$ is odd.

$$\text{var}(X) = E[X^2] = \int_{-\infty}^{\infty} \underbrace{\frac{x^2}{\sqrt{2\sigma^2}} e^{-\sqrt{\frac{2}{\sigma^2}}|x|}}_{\text{even}} dx = 2\int_0^\infty \frac{1}{\sqrt{2\sigma^2}} x^2 e^{-\sqrt{\frac{2}{\sigma^2}}x} dx = \frac{2}{\sqrt{2\sigma^2}} \int_0^\infty \underbrace{x^2}_{u} \underbrace{e^{-\sqrt{\frac{2}{\sigma^2}}x} dx}_{dv}$$

$$= \frac{2}{\sqrt{2\sigma^2}}\left(x^2 \frac{e^{-\sqrt{\frac{2}{\sigma^2}}x}}{-\sqrt{2/\sigma^2}}\Big|_0^\infty\right) - \int_0^\infty \frac{e^{-\sqrt{\frac{2}{\sigma^2}}x}}{-\sqrt{2/\sigma^2}} 2x\,dx$$

$$= \frac{2}{\sqrt{2\sigma^2}} \frac{2}{\sqrt{2/\sigma^2}} \int_0^\infty x \cdot e^{-\sqrt{\frac{2}{\sigma^2}}x} dx = 2 \int_0^\infty \underbrace{x}_{u} \underbrace{e^{-\sqrt{\frac{2}{\sigma^2}}x} dx}_{dv}$$

$$= 2\left[x \frac{e^{-\sqrt{\frac{2}{\sigma^2}}x}}{-\sqrt{2/\sigma^2}} \Big|_0^\infty \right] - \int_0^\infty \frac{e^{-\sqrt{\frac{2}{\sigma^2}}x}}{-\sqrt{2/\sigma^2}} dx = 2 \int_0^\infty \sqrt{\frac{\sigma^2}{2}}\, e^{-\sqrt{\frac{2}{\sigma^2}}x} dx$$

$$= 2\sqrt{\frac{\sigma^2}{2}} \frac{e^{-\sqrt{\frac{2}{\sigma^2}}x}}{-\sqrt{\frac{2}{\sigma^2}}} \Big|_0^\infty = 2\sqrt{\frac{\sigma^2}{2}} \cdot \frac{1}{\sqrt{2/\sigma^2}} = \frac{\sqrt{2\sigma^2}}{\sqrt{2/\sigma^2}} = \sigma^2$$ ☒

[6] An example of a Gaussian mixture PDF is

$$p_X(x) = \frac{1}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2}, \quad -\infty < x < \infty$$

Determine its mean and variance.

Solution:

$$E[X] = \int_{-\infty}^{\infty} x \cdot p_X(x)\, dx = \int_{-\infty}^{\infty} x \left[\frac{1}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2} + \frac{1}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2} \right] dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2}}_{N[1,1]} dx + \frac{1}{2} \int_{-\infty}^{\infty} x \cdot \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2}}_{N[-1,1]} dx$$

$$= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1) = 0$$

$$\text{var}(X) = E[X^2] - (E[X])^2 = E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot p_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} x^2 \frac{1}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2} dx + \int_{-\infty}^{\infty} x^2 \frac{1}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2} dx$$

$$= \frac{1}{2} \underbrace{\int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2} dx}_{E[X^2] \text{ for } N[1,1]} + \frac{1}{2} \underbrace{\int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2} dx}_{E[X^2] \text{ for } N[-1,1]}$$

$$= \frac{1}{2} \cdot \left[(1+1^2) + (1+(-1)^2) \right] = \frac{1}{2}[2+2] = 2$$ ☒

[7] For a Laplacian PDF with $\sigma^2 = 2$, determine all the moments

Solution:

$$E[X^n] = \frac{1}{j^n} \frac{d^n \phi_X(\omega)}{d\omega^n}\Big|_{\omega=0}, \quad \phi_X(\omega) = \frac{1}{\omega^2+1} \text{ for } \sigma^2 = 2.$$

$$\phi_X'(\omega) = \frac{-2\omega}{(\omega^2+1)^2}, \text{ using } \frac{1}{\omega^2+1} = \frac{1}{2j}\left[\frac{1}{\omega-j} - \frac{1}{\omega+j}\right] \text{ gives}$$

$$\phi_X''(\omega) = \frac{2}{2j}(\omega-j)^{-3} - \frac{2}{2j}(\omega+j)^{-3}$$

$$\phi_X'''(\omega) = -\frac{2 \cdot 3}{2j}(\omega-j)^{-4} + \frac{2 \cdot 3}{2j}(\omega+j)^{-4}$$

$$\vdots$$

$$\phi^{(n)}(\omega) = \frac{(-1)^n n!}{2j}(\omega-j)^{-n-1} - \frac{(-1)^n n!}{2j}(\omega+j)^{-n-1}$$

$$\frac{1}{j^n}\phi_x^{(n)}(0) = \frac{1}{j^n}\left[\frac{(-1)^n n!}{2j}(-j)^{-n-1} - \frac{(-1)^n n!}{2j}\,j^{-n-1}\right] = \begin{cases} 0, & n \text{ odd} \\ n!, & n \text{ even} \end{cases}$$

[8]. If $X \sim N[0, \sigma^2]$, determine $E[X^2]$ using the characteristic function.

Solution: $\phi_x(\omega) = e^{-\frac{1}{2}\sigma^2\omega^2}$

$$E[X^2] = \frac{1}{j^2}\left.\frac{d^2\phi_x(\omega)}{d\omega^2}\right|_{\omega=0}$$

$$\frac{d\phi_x(\omega)}{d\omega} = -\sigma^2\omega\, e^{-\frac{1}{2}\sigma^2\omega^2}, \qquad \frac{d^2\phi_x(\omega)}{d\omega^2} = (-\sigma^2\omega)^2\, e^{-\frac{1}{2}\sigma^2\omega^2} - \sigma^2 e^{-\frac{1}{2}\sigma^2\omega^2}$$

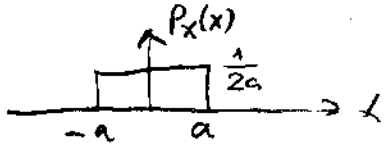
$$E[X^2] = \frac{1}{j^2}\left[(-\sigma^2\omega)^2 e^{-\frac{1}{2}\sigma^2\omega^2} - \sigma^2 e^{-\frac{1}{2}\sigma^2\omega^2}\right]\Big|_{\omega=0} = \frac{1}{j^2}(-\sigma^2) = \sigma^2$$

[9] The probability that a random variable deviates from its mean by an amount of γ in either direction is to be less than or equal to $\frac{1}{2}$. What should γ be?

Solution: $P[|X-E|>\gamma] \le \frac{\text{Var}(x)}{\gamma^2} = \frac{1}{2} \Rightarrow \gamma^2 = 2\,\text{var}(x)$ or $\gamma = \sqrt{2\,\text{var}(x)}$

[10] Determine the probability that $|x|>\gamma$ if X is uniformly distributed over $[-a, a]$. Next, compare these results to the chebyshev bound for $a=2$

Solution:



$$P(|x|>\gamma) = \int_{-\infty}^{-\gamma} P_x(x)\,dx + \int_{\gamma}^{\infty} P_x(x)\,dx$$

$$= \begin{cases} 0, & \text{if } \gamma > a \\ 1-\frac{\gamma}{a}, & \text{if } 0 \le \gamma \le a \end{cases}$$

For $a=2$, $P[|x|>\gamma] = \begin{cases} 0, & \gamma > 2 \\ 1-\frac{\gamma}{2}, & 0 \le \gamma \le 2 \end{cases}$

Chebyshev bound: $P[|X-E(x)|>\gamma] \le \frac{\text{var}(x)}{\gamma^2}$ $\quad (E(x)=0)$

$$\Rightarrow P[|x|>\gamma] \le \frac{\text{var}(x)}{\gamma^2} = \frac{(2a)^2}{12\gamma^2} = \frac{4a^2}{12\gamma^2} = \frac{a^2}{3\gamma^2}$$

For $a=2$, $P(|x|>\gamma) \le \frac{4}{3\gamma^2}$

